

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

THERMODYNAMICS

LECTURE-04 **PHYSICS**

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Topics To Be Covered

- 1 Revision *→ Lec-1 to dec 3*
- 2 Work done by (Adiabatic Process)
- 3 2nd Law of thermodynamic
- 4 Carnot Engine
- 5 Kelvin-Planck Statement

Recap - ①

• Thermodynamics ✓

• System ✓

• Surrounding ✓

• universe ✓

• Intinsic property

• Extinsic property

• Isothermal

• Isobaric

• Isochoric

• Adiabatic

• Reversible

• Irreversible

• Cyclic process

• Temperature

• Thermal equilibrium

• Zeroth law of thermodynamics

Recap from 2nd Lecture

• Heat

$$Q_p = nC_p \Delta T \quad Q+ \quad Q-$$

• Internal Energy

$$U = nC_v \Delta T \quad U+ \quad U-$$

• Work

Expansion $W = +ve$ Compression $W = -ive$

isobaric

$$W = P \Delta V$$

Ideal gas equation

$$PV = nRT$$

$$P \Delta V = nR \Delta T$$



lecture_no. 4

(III) WORK DONE BY Adiabatic process

①

Adiabatic process

⇒ The process in which heat remain constant

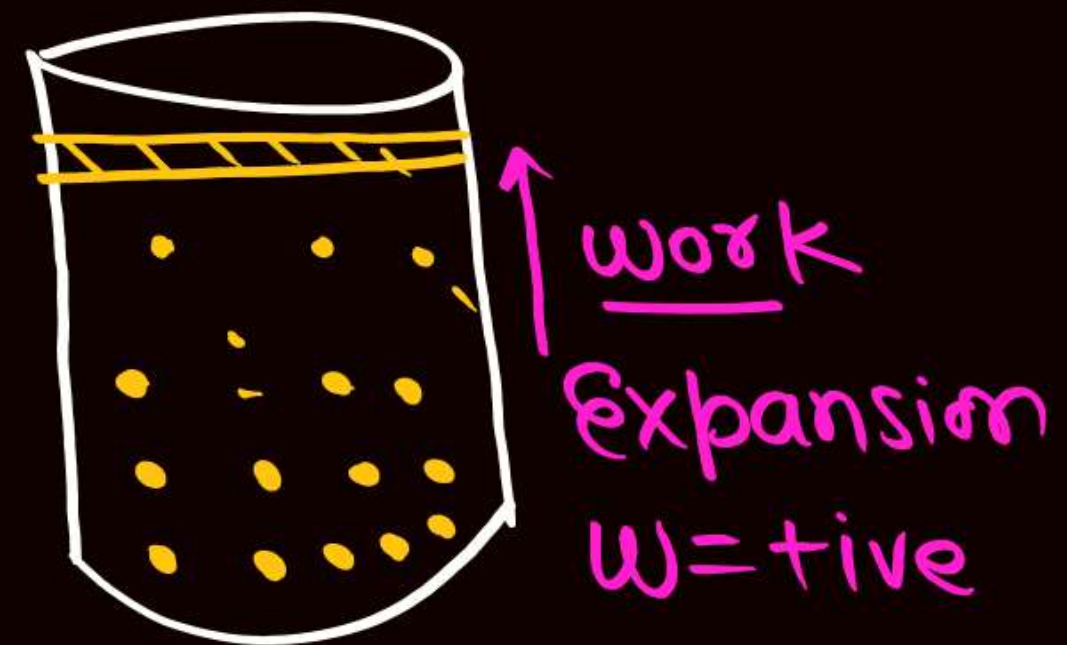
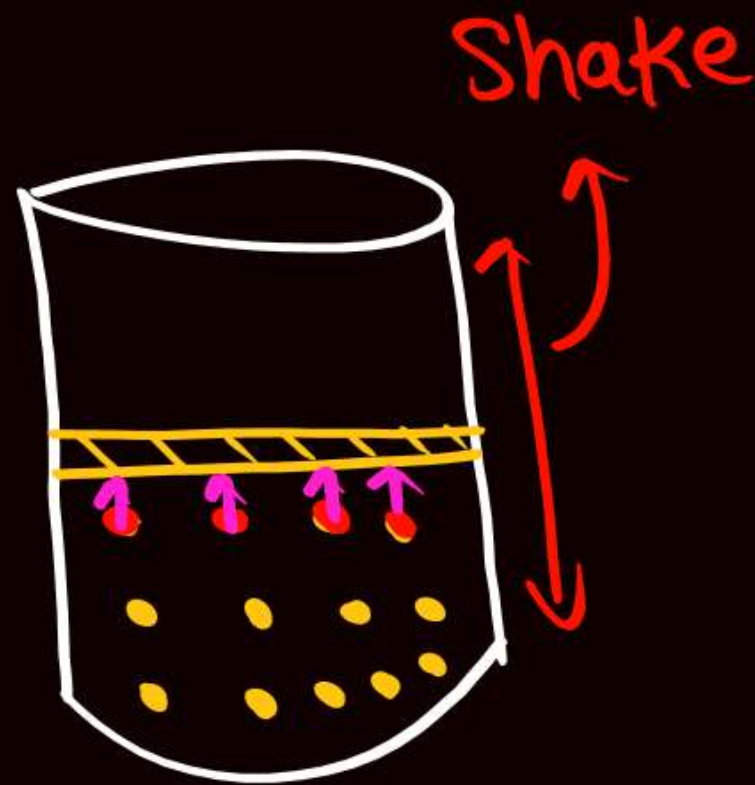
$$Q_i = Q_f = 4J$$

$$\Delta Q = Q_f - Q_i$$

$$= 4 - 4$$

$$= 0$$

$$dQ = 0.$$



maths

1st law of thermodynamics

$$Q = U + W$$

$$\cancel{dQ} = dU + dW$$

$$0 = \underline{dU} + \underline{dW}$$

$$0 = nC_v dT + dW$$

$$-nC_v dT = dW$$

$$\Rightarrow U = Q_v = nC_v \Delta T$$

②

$$\underline{U} = nC_v \underline{\Delta T}$$

$$\rightarrow \underline{dU} = nC_v dT$$

$$dw = -\underline{n}C_vdT \quad n=1$$

$$dw = -1 \times C_vdT$$

$$\underline{dw} = -\underline{C_vdT} \text{ (M B-Point)}$$

on integration b/s

$$\int \underline{dw} = \int_{T_1}^{T_2} -\underline{C_vdT}$$

$$W = -C_v \int_{T_1}^{T_2} dT$$

$$\int dx = x$$

$$\int dT = T$$

③

$$W = -C_v \left[\underline{T} \right]_{T_1}^{T_2}$$

$$W = -C_v (T_2 - T_1)$$

$$W = -C_v T_2 + C_v T_1$$

$$W = C_v T_1 - C_v T_2$$

$$W = \underline{C_v} T_1 - \underline{C_v} T_2$$

Note

$$C_p - C_v = R$$

Divide b/s by C_v

$$\gamma \left(\frac{C_p}{C_v} - \frac{C_v}{C_v} \right) = \frac{R}{C_v}$$

$$(\gamma - 1) = \frac{R}{C_v}$$

Expansion
Adiabatic

→ Compression

Contraction
Adiabatic

$$C_v = \frac{R}{(\gamma - 1)}$$

$$W = \underline{C_v} (T_1 - T_2)$$

$$C_v = \frac{R}{\gamma - 1}$$

$$W = + \frac{R}{(\gamma - 1)} (T_1 - T_2)$$

$$W = - \frac{R}{(\gamma - 1)} (T_1 - T_2)$$

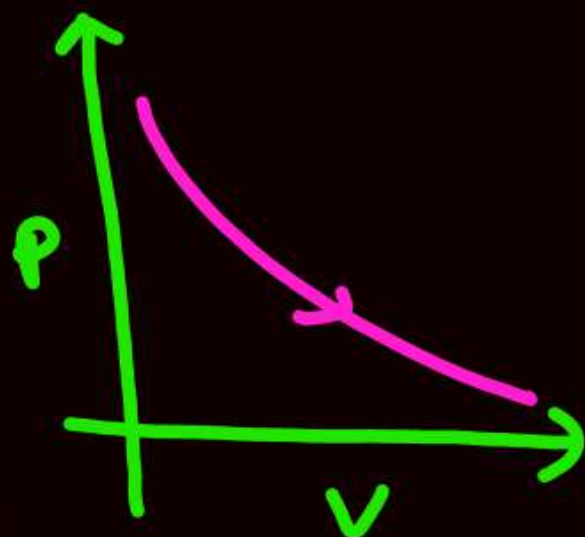
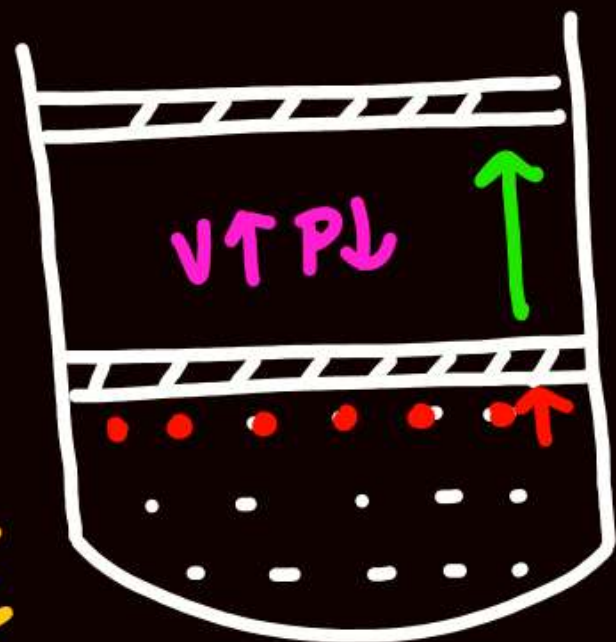
④

PV - for work done by Adiabatic process

Neet
2021

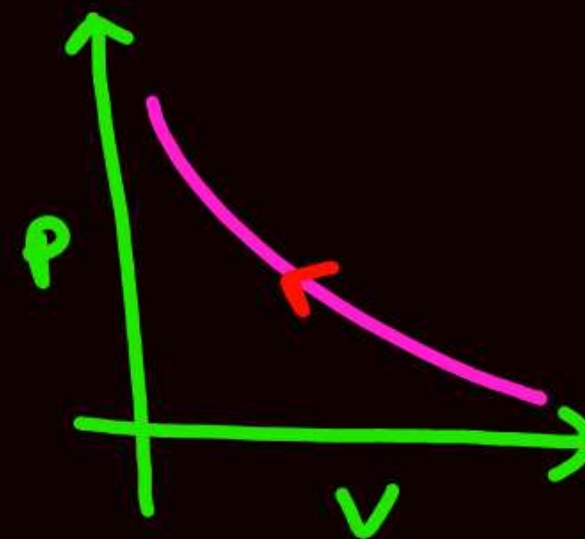
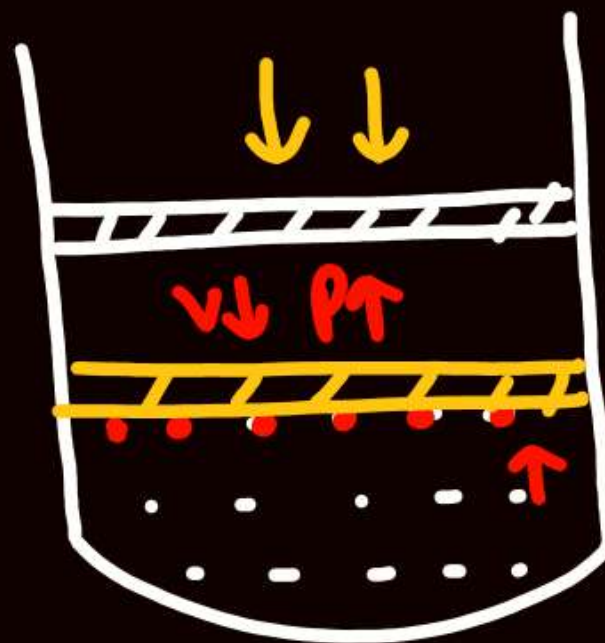
Expansion

$$W = + \frac{R}{(\gamma - 1)} (T_1 - T_2)$$



Compression

$$W = - \frac{R}{(\gamma - 1)} (T_1 - T_2)$$



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(IV) WORK BY Isochoric process

The process in which
Volume Remains constant

$$V_i = V_f = 10$$

$$\Delta V = V_f - V_i$$

$$\Delta V = 10 - 10$$

$$\Delta V = 0$$

$$V_i = V_f$$
$$\Delta V = 0$$



$$W = P \Delta V$$

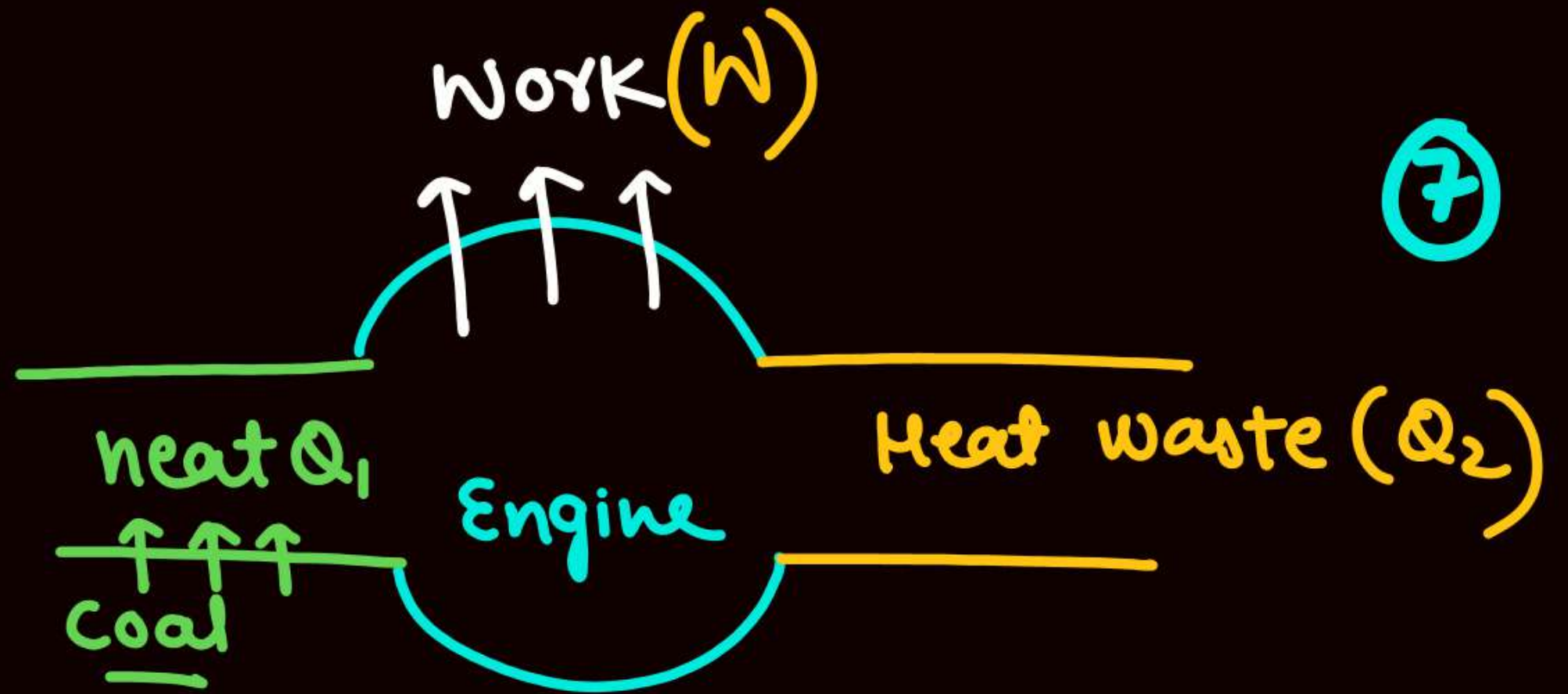
$$W = P \times 0$$

$$W = 0 \text{ J}$$

(6)

Heat engine

It is a device which convert the heat energy into use full work



$$\text{Heat} = \text{Work} + \text{Heat loss}$$

$$Q_1 = W + Q_2$$

100J

70

30

.

Heat engine efficiency $\eta = \frac{W}{Q_1}$

$$= \frac{Q_1 - Q_2}{Q_1}$$

$$\Rightarrow Q_1 = \underline{W} + \underline{Q_2}$$

$$\boxed{Q_1 - Q_2 = W} \leftarrow \text{WORK} \text{ (8)}$$

Heat engine efficiency Percentage

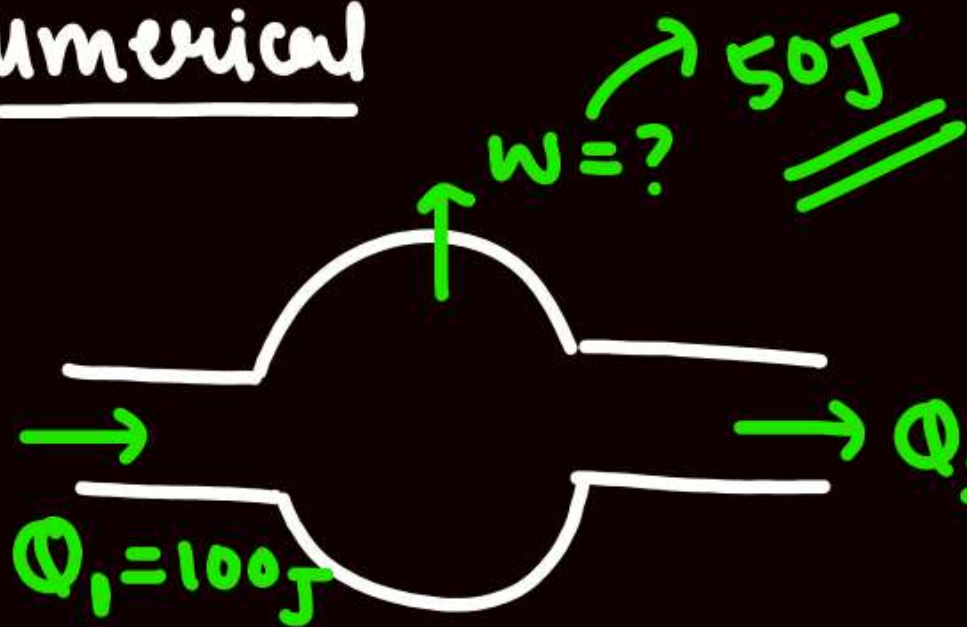
$$\eta = \frac{W}{Q_1} \times 100$$

$$= \frac{Q_1 - Q_2}{Q_1} \times 100.$$

$$\Rightarrow Q = W + Q_2$$

$$\boxed{Q - W = Q_2} \leftarrow \text{heat loss}$$

Numerical



$$(ii) \eta = \frac{W}{Q_1} = \frac{50}{100} = \frac{1}{2} = 0.5$$

9

$$(iii) \eta \% = \left(\frac{W}{Q_1} \right) \times 100$$

$$= 0.5 \times 100 = 50\%$$

Sol (i) Work

$$Q_1 = W + Q_2$$

$$Q_1 - Q_2 = W$$

$$100 - 50 = W$$

$$50J = W$$

Note

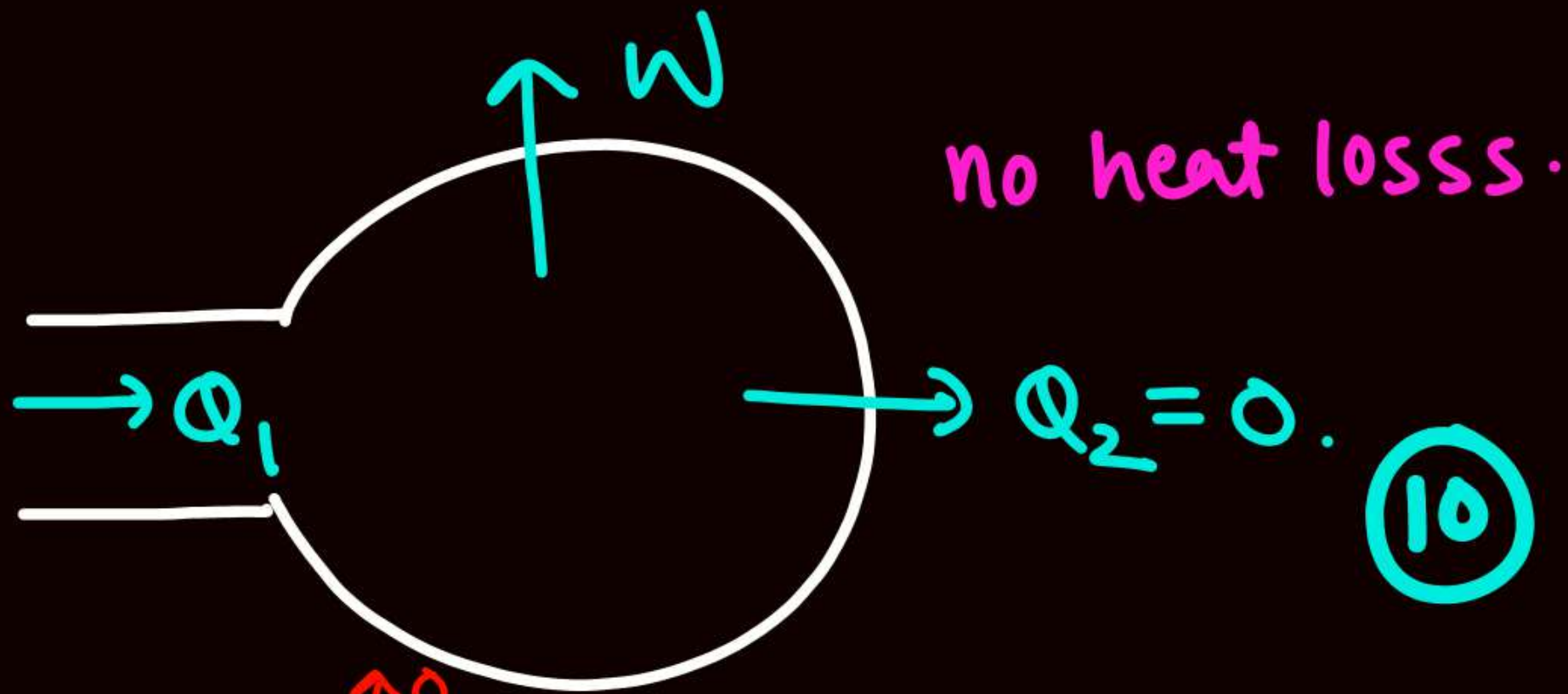
Carnot engine

⇒ ideal engine

⇒ we cannot construct this engine.

⇒ no heat loss here.

$$\Rightarrow \eta\% = 100\%$$



$$\Rightarrow Q_1 = W + Q_2$$

$$Q_1 = W$$

$$\Rightarrow \eta = \frac{W}{Q_1} = \frac{Q_1}{Q_1} = 1$$

$$\Rightarrow \eta\% = \frac{W}{Q_1} \times 100 = 1 \times 100 = 100\%$$

2nd Law of thermodynamic



2nd law of Thermodynamics

S-I
K&P

S-II
C-Statement



(i) Kelvin-Planck Statement (Heat Engine Statement)



It is impossible to construct the engine which convert all heat (Q_1) into useful work (W)

OR

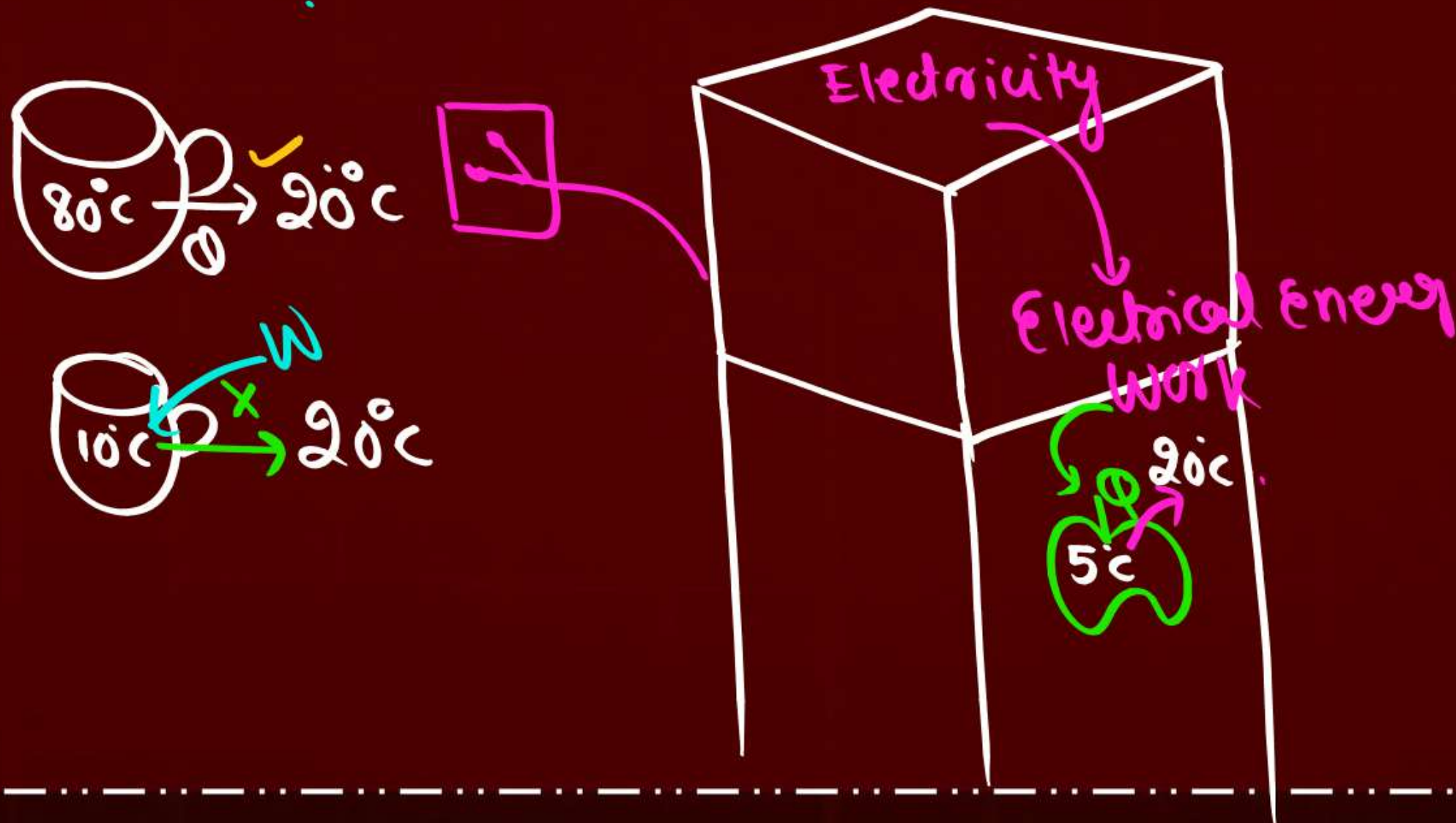
no heat engine can be of 100%.



(ii) Clausius Statement (Refrigerator Statement)

It is impossible to extract the heat from Colder Body

If you want to extract the heat from Colder Body
you have to do the Extra External Work



Question

Aaj Kitna Samjh Aaya?



100%



50%



Zero



Maine khud nhi padha



HOMework



Copy complete
Post

NLM

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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